# DegreeZero — Methodology

Scoring philosophy, formulas, and bias controls · 2026-06-09

This document is the public, versioned methodology. Scores always display the methodology
version that produced them; historical scores are immutable and re-derivable.

## 1. Philosophy: procedural vs. substantive neutrality

Two kinds of neutrality must be separated:

- **Procedural neutrality** — identical rules, code, sources, and process for everyone; fully
transparent, reproducible, auditable. **We target ~100% of this.**
- **Substantive (value) neutrality** — deciding what counts as a *good* outcome. **Impossible**:
aggregating liberty vs. equality vs. growth vs. safety into one number is itself an ideology.

Therefore degreezeor does **not** emit a default normative "good/bad" score. It answers a *factual*
question and pushes values to a user-controlled layer with a neutral default.

> **The factual question:** *Did the measurable outcome tied to this action's own stated objective
> move, relative to a defensible baseline, and how much is credibly attributable to this official —
> with what confidence — and here is every official source?*

### Three irreducible non-empirical residues (labeled, never hidden)

1. **Metric choice** — even "use the stated objective" privileges the sponsor's framing.
2. **Baseline/counterfactual choice** — different defensible designs give different deltas.
3. **Attribution** — causal credit in multi-actor, multi-causal systems is underdetermined.

When a residue dominates → output is **"insufficient evidence / non-scoreable"**, *not* a low score.

## 2. Source hierarchy (provenance tiers)

- **Tier 0** — primary record of the action (Congress.gov / GovInfo / Federal Register).
- **Tier 1** — official statistical outcome data (BLS, BEA, Census, FBI, CDC, EIA, USAspending…).
- **Tier 2** — official nonpartisan analysis (CBO, GAO, CRS, agency IGs).
- **Tier 3** — convenience mirrors (FRED, OpenStates) — only with verification back to Tier 0/1.

Every stored datum carries source_url, content_hash, retrieved_at, tier, and a native identifier (e.g., Public Law number, BLS series id).

## 3. Scoreable unit

- Atomic data unit: a polymorphic **Action** (vote / bill / law / EO / regulation / budget / …).
- Atomic **scoreable** unit: the tuple **(Action → Objective → Metric → Baseline → Outcome)**, an

*Evaluation Unit (EU)*.
- Outcomes attach to **implemented policies**, not votes. Votes/sponsorship/signature are

**attribution channels** that propagate a policy's measured outcome to officials.

## 4. Outcome score (sign-neutral)

```
delta             = observed − baseline_pooled
delta_toward_goal = sign_goal · delta          # sign_goal fixed at pre-registration from the objective
z                 = delta_toward_goal / noise_scale
S_outcome         = 100 · logistic(1.702 · z)    # 50 = no effect
```

sign_goal = +1 if a *rise* in the metric moves toward the stated goal, −1 if a *fall* does. This is the ideology-free core: we only ask whether the metric moved toward the goal **you stated**.

## 5. Baseline ensemble & model dependence

A baseline ensemble estimates the counterfactual. Methods plug in behind one BaselineMethod interface:

- pretrend_projection — project the pre-period linear trend to the evaluation point. The formal

answer to *"don't credit inherited trends"*: the pre-existing trajectory **is** the baseline.
- flat_last_value — naive no-change counterfactual.
- difference_in_differences — counterfactual = treated pre-level + the control group's pre→post

change; eligible only when pre-trends are parallel. Differences out shocks common to treated and

control units.
- synthetic_control — counterfactual = a nonnegative, sum-to-one weighted blend of donor units

that best matches the treated unit's pre-period path (Abadie-style); eligible only when the pre-fit is tight. Strongest identification in the system.

**Tiered pooling:** when a comparison design (DiD / synthetic control) is eligible it *drives* the estimate; naive single-series baselines are reported for transparency but never dilute a well-identified counterfactual. Comparison designs require donor/control units (sub-national variation) — e.g. a state policy scored against a pool of donor states' official series.

model_dependence ∈ [0,1] rises with sign disagreement / spread across the active methods and lowers
confidence. A bootstrap (deterministic seed) yields a 95% CI on the delta.

## 6. Attribution

attribution = f(formal_authority, pivotality, implementation_control), reported as an interval,
**always** leaving a large explicit unattributable_residual (Σ humans + residual = 1,
human total capped at 0.70). Vote pivotality ≈ 1/(margin+1) — high only for razor-thin votes.
For an enacted law, the final-passage roll-call of **both chambers** (House clerk XML + Senate
LIS XML, both Tier-0) is ingested, and each chamber's winning-side members receive decisive-vote
attribution scaled by **that chamber's own margin**. (Senate records key on lis_member_id; the
unitedstates/congress-legislators dataset bridges it to Bioguide — used strictly for identifier resolution, never as a scored quantity.)

## 7. Confidence gate (prevents false precision)

```
C = c_design · c_data · c_attrib · c_modeldep · c_sensitivity   # each ∈ [0,1]
```

- c_design — identification strength. A single federal time series with a pre-trend baseline is
capped (it cannot separate a policy from concurrent macro shocks), so federal macro cases often
fall below the publish threshold **by design**.
- c_data — provenance tier + completeness.
- c_attrib — 1 − mean(attribution interval width).
- c_modeldep — 1 − model_dependence.
- c_sensitivity — 0.5 when the effect's direction flips across defensible lag horizons
(fragile), else 1.0. Ties the sensitivity analysis (§9.10) back into the score.

If C < 0.60 (default), the composite is **suppressed** and the EU renders **"insufficient** evidence." **This is the correct, humble behavior — not a failure.**

## 7b. Two evaluation modes

- **Baseline-relative (causal effect):** "did the metric move vs. a defensible counterfactual?"

(pre/post, DiD, synthetic control). Identification is the hard part.

- **Target-relative (promise-keeping):** "did the policy deliver its own pre-registered, source-linked
numeric target?" Realized = official data; target = the policy's CBO/statutory number.
**Integrity guardrail:** target-relative earns high identification (declared_target_direct, 0.90)
**only when the realized series is directly attributable to the action** (e.g. a law's own
DEFC-tagged USAspending total). Economy-wide realized series are declared_target_confounded
(0.35) and stay gated. This is how target-relative scoring reduces "insufficient evidence"
*legitimately* — it only scores what's genuinely attributable.

## 8. Default output vs. opt-in composite

- **Default (public):** the decomposed factual component vector — outcome, evidence, attribution,
alignment, dataquality, durability — each with uncertainty, plus confidence and the source trail.
- **Composite (opt-in, value-laden):** EU_score = gate(C) · Σ_k w_k · S_k. The **neutral default**
averages only the **achievement** components — outcome and durability, both directional toward
the stated goal — and then scales by confidence C. Crucially, evidence (=c_design) and
dataquality (=c_data) are *already inside* C, so averaging them into the composite too would
**double-count confidence**; alignment is a metric-fidelity quality measure shown for context.
durability is goal-directional, so a *persistent move away from the objective* correctly scores
**low** (not high). Value-laden lenses (cost, distribution) are off by default. Users may
reweight any component; the UI watermarks non-neutral weightings.
- **Official roll-up:** attribution-weighted mean of EU scores, always reported with **coverage**
(fraction of actions that are scoreable) and confidence.

## 8b. Two layers: "did it work?" vs. "what did they act on?"

A scoreable unit requires an isolatable, measured outcome (§3). Most legislative behaviour does
not meet that bar, yet voters reasonably ask *what an official acted on*. We therefore publish two
clearly separated layers, never blended:

- **Scored layer ("did it work?"):** composite scores and the factual component vector above. Only

EUs that pass the confidence gate (§7) receive a composite.

- **Record layer ("what they acted on"):** the **unscored** legislative record — bills a member **sponsored** and **cosponsored** (Congress.gov), plus how they **voted** on recorded roll-call votes (House: clerk.house.gov; Senate: senate.gov — keyless Tier 0, senators resolved via the lis↔bioguide crosswalk), and for the executive, the **executive orders a president** signed **(Federal Register) — grouped by the objective category taxonomy. These are** source-anchored facts describing breadth and positions by topic. They are **never** fed into any score, attribution, or the party-symmetry monitor; they exist purely so the public record of activity is visible alongside (and visually distinct from) the rigorous scored layer. The per-official voting record uses only comprehensive roll-calls (so it never double-counts the attribution-only passage votes that feed the *scored* layer).

The record layer is ingested recency-first and incrementally, capped per run so the per-key request budget is respected; repeated nightly runs fill in the remaining history.

## 9. Bias-minimization protocol

1. Identical pipeline for all officials; **scoring code is party-blind** — enforced statically (no scoring/ module reads party) **and** behaviourally (swapping an official's party cannot change a stored score).
2. **Pre-registration**: metric + baseline + lag + sign_goal hashed to the audit chain **before** outcome data is fetched — kills outcome-driven cherry-picking.
3. **Party-masked** objective→metric selection.
4. No editorial labels; numbers, intervals, and sources only.
5. Open methodology + open-source formula; changes via reviewed PRs.
6. Uncertainty always shown; composite gated by confidence.
7. Append-only, hash-chained audit log; pinned, bit-reproducible score runs.
8. Always a large unattributable residual.
9. **Sensitivity analysis** — each scoreable EU re-evaluates its outcome across defensible lag horizons (12–60 months) and reports robustness (direction stability + significance fraction); a sign that flips with the horizon is flagged as not-robust. (/api/evaluation-units/{id}/sensitivity.)
10. **Dispute / appeal process** — anyone can trigger an independent, deterministic re-run; the result (reproduced vs. corrected) + public diff is recorded on the audit chain.
11. **Relationship graph** — officials↔actions↔jurisdictions↔metrics exposed for transparency.

12. **Reproducibility self-audit** — every published score can be independently re-derived from
its stored inputs + pinned methodology and must reproduce its hash bit-for-bit; the platform-wide
audit re-runs them all and flags any mismatch (non-determinism / tampering) as a hard failure.
(/api/integrity/reproducibility, degreezeor verify-scores.)
13. **Integrity-at-scale monitoring** — because even a party-blind formula can produce uneven
*distributions*, the platform publishes the party-level distribution of scored outcomes
(attribution-weighted mean composite + coverage per party) and flags any systematic
composite-gap or scored-share-gap for **human methodological review** of metric/baseline
choices — never an automated correction, and never a change to any individual score. Party is
read here for audit only. (/api/integrity/party-symmetry, degreezeor party-symmetry, and the
"Integrity" view.)
14. **Display neutrality** — party is removed from the entire user-facing experience (no party
label, filter, or sort on officials); only the name and, where derivable from the public record,
the office (President, Governor, Senator, Representative) are shown. Party remains in the data
layer solely for the audit-only party-symmetry monitor (item 13). Score-bearing surfaces use a
neutral mauve tone, never green-as-good, red-as-bad, or party-blue. Actions are grouped into
objective topic **categories** (jobs and economy, cost and spending, health, public safety,
energy and environment, poverty and income, education) derived deterministically from each
action's official subject domain and the metric it was measured against; categories never feed
scoring. Official-to-official and action context is **descriptive only** (how a result sits next
to the typical scored result for its action type or category, shown only with a sufficient
sample), never a ranking.

## 10. What cannot be made fully empirical

Metric selection, baseline choice, causal attribution, and *whether an outcome is desirable*. These
are surfaced explicitly (intervals, model-dependence, residual, value-weight layer), never hidden.

## 11. "Insufficient evidence" is triggered by

Missing/contradictory data, no operational metric, no implementation, dominant external shocks,
sign disagreement across baselines, or confidence below the publish threshold.

## Roadmap

- **Phase 1 (shipped):** US Congress, enacted economic/fiscal laws; Congress.gov + CBO links +
BLS; pre-trend/flat baselines; sponsor/signer/decisive-vote attribution (House + Senate roll-calls);
confidence gate; API + UI.
- **Phase 2 (in progress): difference-in-differences & synthetic control shipped** (with
tiered pooling and state-policy scoring on official BLS state series — e.g. the Kansas 2012 tax
cuts demo, which clears the gate and yields a real composite; the curated state set now also
includes North Carolina 2013, Wisconsin 2011, Maine 2011, Indiana 2013, Ohio 2013, and Missouri
2014, each source-verified, with the synthetic-control pre-fit gate deciding which clear the gate
vs. honestly abstain). The comparison-design scorer also runs on **state wage** (BLS average
hourly earnings; minimum-wage laws in California, Massachusetts, Maryland), **state poverty/income**
(Census SAIPE; e.g. the California EITC), **state health coverage** (Census SAHIE uninsured rate;
e.g. California's ACA Medicaid expansion, which clears the gate), and **state emissions** (EIA
state CO2; e.g. California SB 350) — each measured against the policy's own stated goal with the
pre-registered sign (more jobs/wages/income, or less poverty/uninsured/CO2). The curated set spans
the most populous states first and both parties (for example Michigan 2011 business tax, New Jersey
2019 minimum wage, Virginia 2020 and Washington 2019 clean-energy laws), with the gate deciding
which clear vs. abstain. **Executive orders shipped**
(Federal Register Tier-0 ingestion; the signing president carries high unilateral executive
authority vs. shared law-signing). **Final agency rules (regulations) shipped** (Federal Register
Tier-0; attributed to the administration in office on the rule's effective date). **Senate**
roll-calls shipped **(both chambers' passage voters credited, per-chamber pivotality).** Budget
execution **scored for every toptier agency (reliable, commensurable by**

**construction). Remaining:**

more domains (energy, health, education); deeper regulatory outcome metrics.

- **Phase 3:** state/local government; distributional & cost lenses; user value-weight profiles;

reproducible notebooks; third-party audits.

- **Ultimate:** all levels of government, real-time ingestion, full causal ensemble, audit program.